Stability of Halide Perovskite Solar Cells

A panorama of seven coupled degradation channels — and why ion migration sits at the centre

146	50%	2000–2026	7	13
papers	since 2020	span	themes	cross-cutting

The intrinsic-instability paradox. The soft, ionic, low-formation-energy lattice that gives perovskites their excellent optoelectronics is the same lattice that permits facile ion migration, low-barrier phase transitions, and chemical attack. Instability cannot be eliminated, only suppressed. Across all seven channels below, **ion migration is the mechanistic hub**: a ~0.2 eV rise in its activation energy cuts the migration rate ~1000× at room temperature — the largest single lever on operational stability, and the target of the dopant-screening programme this survey supports.

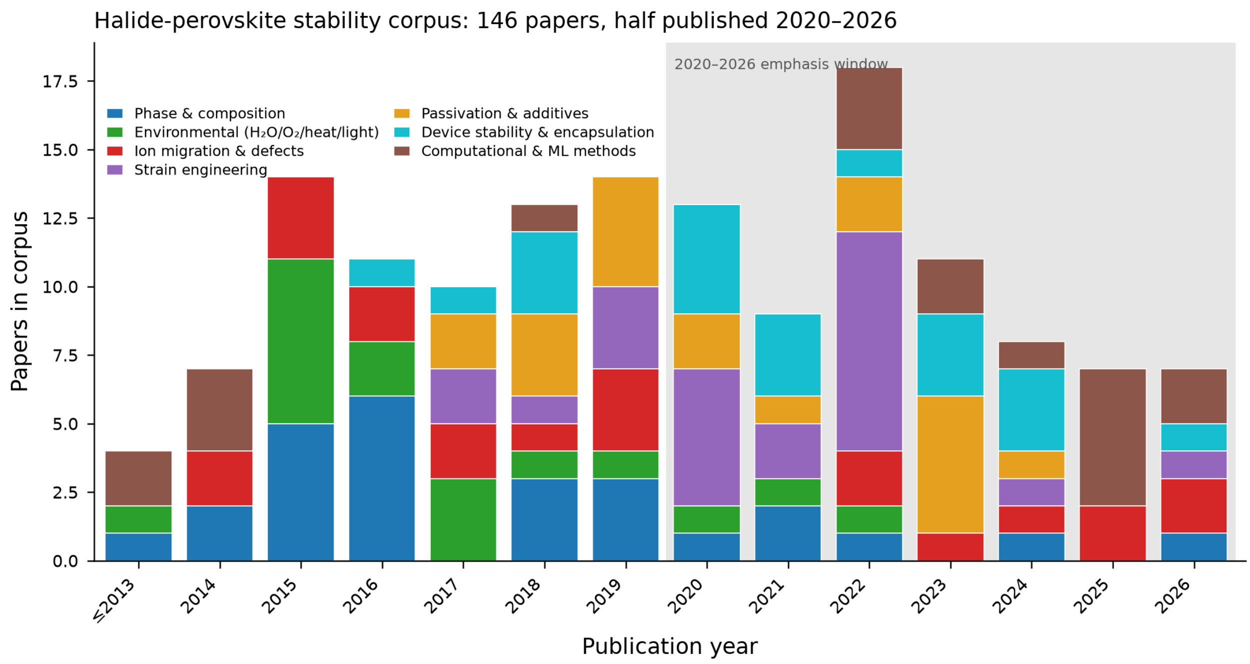

Fig. 1 — Publication timeline of the 146-paper corpus, stacked by theme; shaded = 2020–2026 emphasis window.

The thematic landscape

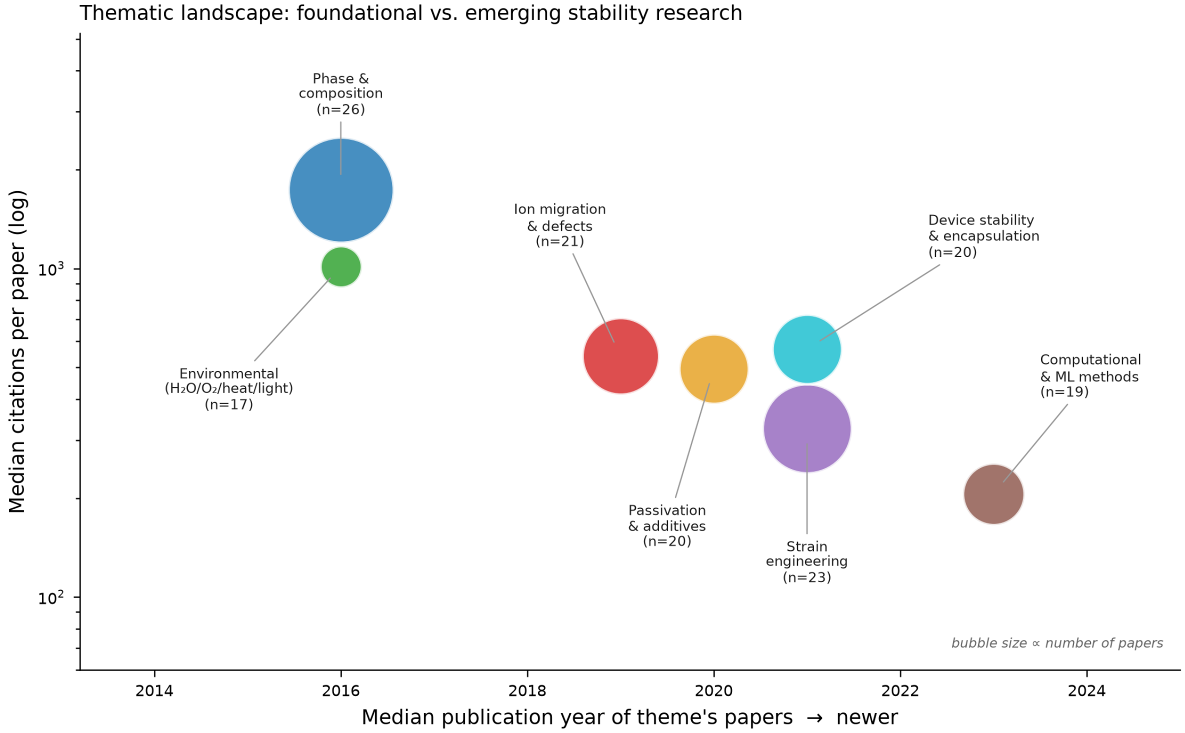

Fig. 2 — Each theme by median publication year (x) and median citations (y, log); bubble area ∝ paper count. Foundational pillars (phase, environmental) upper-left; emerging fronts (computational, strain) lower-right.

The seven stability channels

Each panel: the channel's core finding, distilled. Landmark papers in parentheses; full DOIs in the accompanying review and CSV.

§3.1 Phase & composition

- Black perovskite phases ($FAPbI_3$, $CsPbI_3$) are thermodynamically metastable at operating temperatures, spontaneously converting to wide-gap inactive polymorphs unless kinetically trapped (Li et al. 2008).
- Compositional alloying (Cs/FA/MA across A-site and I/Br across halide) tunes tolerance factor toward perovskite-forming window's stable center while raising demixing barriers (Li et al. 2015).
- $FAPbI_3$ black phase achieves certified >25% efficiency via pseudo-halide (formate) anion engineering, suppressing deep traps while enabling phase-pure growth without Br or Cs widening (Jeong et al. 2021).
- $CsPbI_3$ nanocrystals (quantum dots) exploit surface-energy effects at nanoscale to stabilize cubic α-phase at room temperature and maintain efficient photovoltaics (Swarnkar et al. 2016).

§3.2 Environmental (H_2O / O_2 / heat / light)

- Halide perovskite degradation is a set of chemically distinct defect-mediated channels—hydrolysis, photo-oxidation, thermal decomposition, and photocatalysis—each amplified by mutual interaction (Khenkin et al. 2020).
- Water attack on $MAPbI_3$ is a reversible kinetic process forming monohydrate and dihydrate intermediates before irreversible conversion to PbI_2, accelerating at grain boundaries (Leguy et al. 2015).
- Photo-oxidation via superoxide formation is defect-controlled and the primary operational bottleneck; light + oxygen + trace water degrade perovskites synergistically faster than pairwise combinations (Siegler et al. 2022).
- $MAPbI_3$ thermally decomposes near 85 °C in inert atmosphere independent of moisture and oxygen, and formamidinium successors remain vulnerable to cation volatilization pathways (Juárez-Pérez et al. 2019).

§3.3 Ion migration & defects

- Iodide vacancy migration is the lowest-energy ion transport channel in halide perovskites, with activation energies of 0.1–0.6 eV, driving hysteresis and halide segregation (Eames 2015).
- Self-compensating point-defect pairs pin the Fermi level and maintain defect tolerance while enabling ion mobility, controlled by growth stoichiometry (Walsh 2014).
- Current–voltage hysteresis timescale (sub-second to seconds) directly reports ionic redistribution under bias and quantifies migrating-species density and mobility (Snaith 2014).
- Grain size and morphology, not intrinsic bulk barriers, dominate effective ion diffusivity in polycrystalline perovskites, explaining order-of-magnitude scatter in reported barriers (Ghasemi).

§3.4 Strain engineering

- Tensile strain weakens Pb–X bonds and lowers defect-formation and ion-migration activation energies in halide perovskites (Yang et al. 2022).
- Compressive strain raises ion-migration barriers and halide-segregation activation energy, stabilizing mixed-halide films for >1000 h at 85 °C (Xue et al. 2020).
- Residual tensile strain arises from thermal-expansion mismatch during cooling and compositional gradients; light soaking induces reversible lattice expansion (Zhu et al. 2019).
- Strain engineering at charge-transport interfaces is now front-line stability strategy, addressing real-world failure modes encapsulation cannot defeat.

§3.5 Passivation & additives

- Halide vacancy is the pivotal defect controlling both non-radiative recombination and ion migration; passivation targets it simultaneously for efficiency and stability (Chen et al. 2019).
- Ammonium-halide surface treatments hydrogen-bond to surface sites and quench vacancy-derived states; high-pK_a ammonium cations resist deprotonation for durable thermal stability (Wang et al. 2023).
- 2D organic-halide capping layers passivate surface defects, suppress halide migration, and block moisture ingress; one-year stable operation demonstrated with aminovaleric-acid 2D/3D interface (Grancini et al. 2017).
- Trace multivalent dopants and alkali-metal additives (K, Rb) immobilize halide at grain boundaries and raise vacancy-migration barriers without compromising optoelectronic quality (Zhang et al. 2023).

§3.6 Device stability & encapsulation

- ISOS consensus protocols transformed incomparable stability claims into standardized tests, enabling cross-lab comparison and long-term meta-analysis (Khenkin et al. 2020).
- Mobile ions screen the device's built-in field under operational load, causing power loss distinct from bulk recombination and dominating operational degradation (Thiesbrummel et al. 2024).
- Gold electrode migration through hole-transport layers causes irreversible thermal degradation invisible in shelf tests; diffusion barriers markedly improve stability (Domanski et al. 2016).
- Dopants in spiro-OMeTAD and PTAA are hygroscopic and mobile; interface engineering alone enabled unencapsulated cells to exceed 1,000 operational hours (Rombach et al. 2021).

§3.7 Computational & ML methods

- Iodide-vacancy migration barrier of ~0.6 eV in $MAPbI_3$ identified as dominant ionic channel and source of current–voltage hysteresis (Eames 2015)
- Machine-learned interatomic potentials (MLIP) now span periodic table but systematically soften energy surfaces, underestimating migration barriers without per-system fine-tuning
- E(3)-equivariant message passing (NequIP, MACE) reduces training data by 1–2 orders of magnitude while achieving DFT-level accuracy
- Foundation models like MACE-MP-0 enable finite-temperature MD across chemistries including perovskites without composition-specific training

What the channels share

Cross-cutting themes

1. Ion migration is the shared substrate, not a separate channel: strain, composition and passivation all earn their stability benefit by raising the migration barrier or removing mobile vacancies.
2. The field's centre of gravity moved from shelf stability (dark storage) to operational stability under simultaneous light, bias and heat — the two correlate weakly.
3. Characterisation now resolves microscopic origins of macroscopic failure: transient ion-drift, multiscale diffusion, and machine-learned force fields at grain boundaries.
4. Computation shifted from rationalising experiments (DFT, mid-2010s) to screening candidates ahead of synthesis (fine-tuned MLIPs, 2020s).

Where the map is still blank

Open questions & research gaps

1. Bulk vs film: the cleanly-computed single-crystal barrier is a lower bound; grain boundaries carry the real device transport.
2. Charge state & long-range electrostatics remain hard for charge-agnostic foundation MLIPs (V_I^{+} vs V_I^{0} differ ~10× in mobility).
3. Most empirical successes still lack mechanistic labels — which defect is immobilised is usually unknown.
4. Lab-to-field gap persists; lead-free / Sn–Pb tandems add Sn^{2+} oxidation as a largely orthogonal channel.

Bottom line. Perovskite instability is the coupled expression of one soft, ionic lattice — not a list of independent faults. The channels connect most tightly through ion migration, the only one encapsulation cannot exclude. Raising the migration activation energy by rational dopant design therefore addresses not one corner of the stability problem but its centre.